

Palak Mishra

PhDScholar, Department of Computer Science & Engg.

Indraprastha Institute of Information Technology Delhi

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Country of Origin and Residence: India



Research Interest

Systems and networks for distributed AI/ML workloads, collective communication performance optimization, edge-cloud architectures, transport protocols for HPC/AI applications.

Education

- **PhD, Computer Science Engineering**
IIIT Delhi | 2025-Present | 9.4 SGPA
Lab- NEST(Network & Edge Systems) | Advisor- Dr. Rinku Shah
 - **B.Tech, Computer Science & Engineering**
Galgotias University | 2019 – 2023 | 8.2 CGPA
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Publications-

MemNetAI at ACM SIGCOMM (2026)

Title: Enabling Hardware-Software Co-Design for Large Scale AI Systems via Fine-Grained GPU Memory Tracing

Advisor- Dr. Rinku Shah, In Collaboration with Marvell Technology

Presented at MemNetAI workshop at ACM SIGCOMM, Denver 2026

IEEE Conference (2023)

- **Title:** “Object Detection Using Convolutional Neural Networks”
 - *Co-authored with and under the guidance of Dr. J.N. Singh (Galgotias University)*
 - Presented at IEEE Conference, 2023
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Technical Skills

- **Programming:** c, c++, Python, Java, JavaScript, Node.js
 - **Frameworks & Systems :** NCCL, CUDA, CUPTI, NSYS, NCU, Linux, eBPF, eGPU, Springboot, Networking Protocols, SASS/PTX
 - **Tools & Platforms:** Git, Github, Docker, Kubernetes, AWS, GCP, Chakra
 - **Databases:** PostgreSQL, MongoDB, MySQL
 - **Graduate Subjects :** Networks for AIML Systems, Graduate Algorithm, Graduate Systems, IP, IS
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Projects : [GitHub](#)

Network Stack & System Tracing with eBPF for Containerisation and Non-Containerisation

Research Project

On a testbed of 2 1xRTX GPU laptops and a server with 2 H100 GPUs, executed DDP on ResNet, and profiled the network stack, CPU, and GPU(GPU profiling done using function call metadata) with and without containerisation to test its overhead cost.

Low Level Decomposition of Collective primitives

Research Project

Comparative study of collective communication libraries including NCCL, MPI collectives and collective optimizations like UCCL. Focusing on NCCL I dove deeper into how NCCL works on low level using profilers like Nsight compute, Nsight systems, and CUPTI.

Advisor - Dr. Rinku Shah

(August 2025-Present)

Inference of MoE

Research Project

Looking at optimization techniques for MoE models on inference side. Models- Deepseek, KiMi K2, Llama, Qwen.

Programmable Proxy

Research Project

A network proxy to reduce latency in a dynamic network between server and client.

Full Stack Polling App

A software for enthusiasts to conduct polls, analyse the voting patterns, create profiles and showcase the results customised on their chosen fields. With authentication and security features.

Full Stack Note Apps

Typescript full stack app for creating customised notes.

Transaction App

Java springboot app, made to work with mojaloop. Part of Mifos initiative open source project.

Object Recognition App

Mobile app for identifying currency, objects, and voice to text features for emailing.

Professional Experience

- **Reviewer MLSys 2026** Ninth Annual Conference on Machine Learning and Systems (MLSys'26 Artifact Evaluation) Reviewed 2 papers and evaluated the attached artifacts on Chameleon test beds.
- **Department of Computer Science & Engg., IIT Delhi**
Teaching Assistant (Aug 2025 to Dec 2025)
Worked as Teaching Assistant for the course Operating Systems. The work involves grading answer sheets, providing tutorials, and remedial lessons to students.
- **Developer, Birlasoft (2023–2025)** – Worked on scalable backend & data architecture projects; part of Hewlett Packard team.
- **Open Source Contributor, The Mifos Initiative (C4GT, 2024)** – Developed Open Banking fintech app (Java Spring Boot, PostgreSQL). Collaborated with teams internationally.
- **.NET Developer, Cognizant (2023)** – Built multithreaded ASP.NET project; served as Cohort Lead during training.
- **IIT Allahabad(2020)** - Summer Intern, ML Decision Tree Algorithm and Design.

Awards and Achievements

- Smart India Hackathon 2022
 - DSA Certification NPTEL
 - CCNA Certification Cisco
 - Student Body - Club Head
 - CAT 86%ile
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References

- Dr. Rinku Shah
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